21-242 Recitation 12

RREF — Q2. (A)=GA for some invertible G

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that $\mathrm{RREF}(A) = GA$ for some invertible G.

Course-level solution

Each elementary row operation is left-multiplication by an elementary matrix E (swap: a transposition matrix; scale row i by $\lambda \neq 0$: the diagonal matrix with λ in position i; add λ times row j to row i: $I + \lambda E_{ij}$). Every elementary matrix is invertible (the inverse operation is elementary). The algorithm applies finitely many such operations, so $\mathrm{RREF}(A) = E_k \cdots E_1 A = GA$ with $G = E_k \cdots E_1$ invertible.